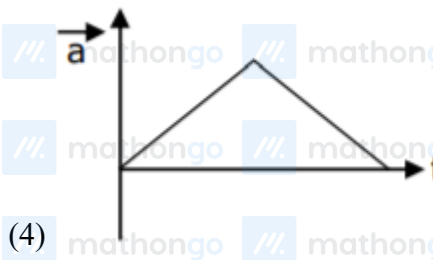
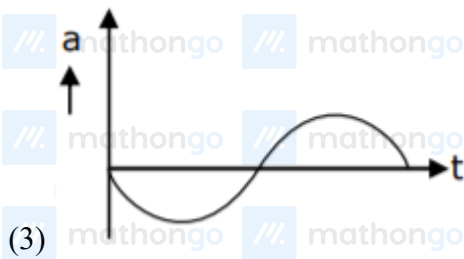
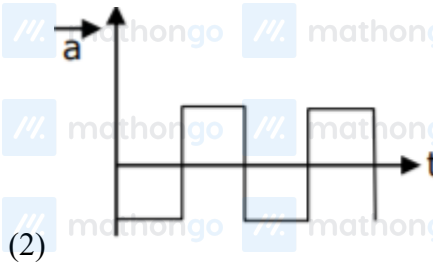
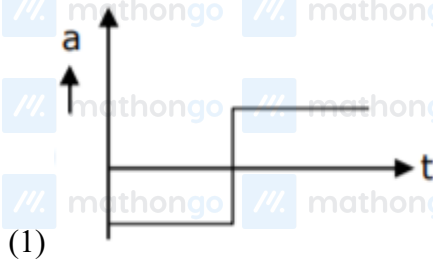
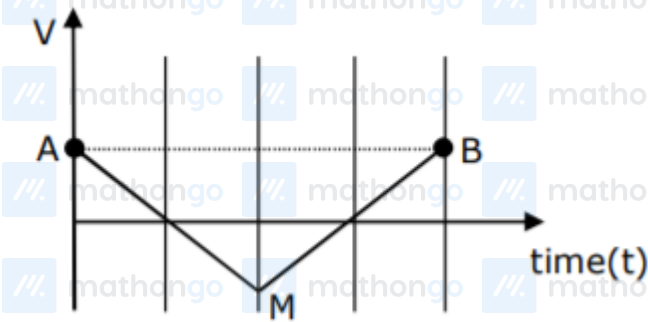


Q1: 24 Feb (Shift 1) - Single Correct

If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph?



Q2: 25 Feb (Shift 1) - Single Correct

Questions with Answer Keys

MathonGo

An engine of a train, moving with uniform acceleration, passes the signal post with velocity u and the last compartment with velocity v . The velocity with which middle point of the train passes the signal post is :

(1) $\sqrt{\frac{v^2 - u^2}{2}}$

(2) $\frac{v - u}{2}$

(3) $\sqrt{\frac{v^2 + u^2}{2}}$

(4) $\frac{u + v}{2}$

Q3: 25 Feb (Shift 2) - Single Correct

A stone is dropped from the top of a building. When it crosses a point 5 m below the top, another stone starts to fall from a point 25 m below the top, Both stones reach the bottom of building simultaneously. The height of the building is:

(1) 45 m

(2) 35 m

(3) 25 m

(4) 50 m

Q4: 26 Feb (Shift 1) - Numerical

A person standing on a spring balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be ____ N. $[g = 10 \text{ m/s}^2]$

Q5: 26 Feb (Shift 2) - Single Correct

A scooter accelerates from rest for time t_1 at constant rate a_1 and then retards at constant rate a_2 for time t_2 and comes to rest. The correct value of $\frac{t_1}{t_2}$ will be:

(1) $\frac{a_1 + a_2}{a_2}$

(2) $\frac{a_2}{a_1}$

(3) $\frac{a_1 + a_2}{a_1}$

(4) $\frac{a_1}{a_2}$

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Answer Key

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Q1 (1)

Q2 (3)

Q3 (1)

Q4 (492)

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

Q5 (2)

MathonGo MathonGo MathonGo MathonGo MathonGo MathonGo

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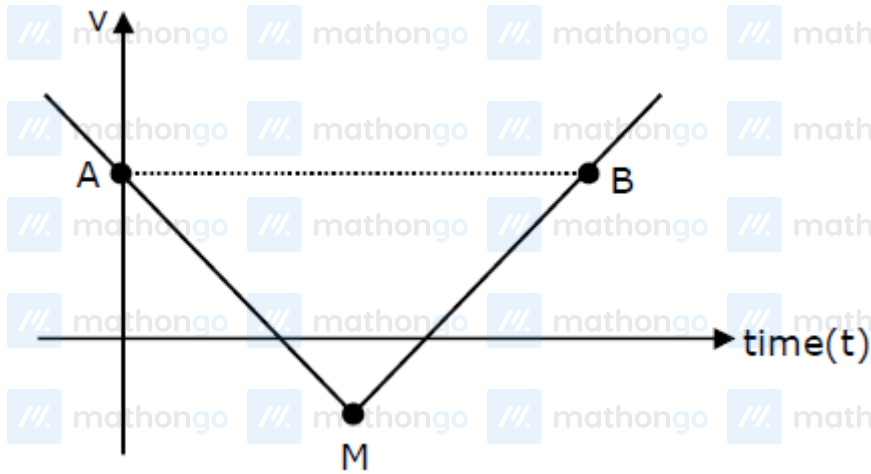
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Q1 (1)



$$a = \frac{dv}{dt} = \text{slope of } (v - t) \text{ curve}$$

If $m = +ve$, then equation of straight line is

$$y = mx + c \Rightarrow v = mt + c$$

(for MB)

If $m = -ve$, then equation of straight line is

$$y = -mx + c \Rightarrow v = -mt + c \text{ (for AM)}$$

If we differentiate equation (1) and (2), we get

$$a_{MB} = +ve = m$$

$$a_{AM} = -ve = -m, \text{ so graph of } (a - t) \text{ will be}$$



Q2 (3)



a = uniform acceleration

u = velocity of first compartment v = velocity of last compartment l = length of train $v^2 = u^2 + 2al$ as (3rd

equation of motion) $v^2 = u^2 + 2al$ (1)

$$v_{\text{middle}}^2 = u^2 + 2a \frac{l}{2}$$

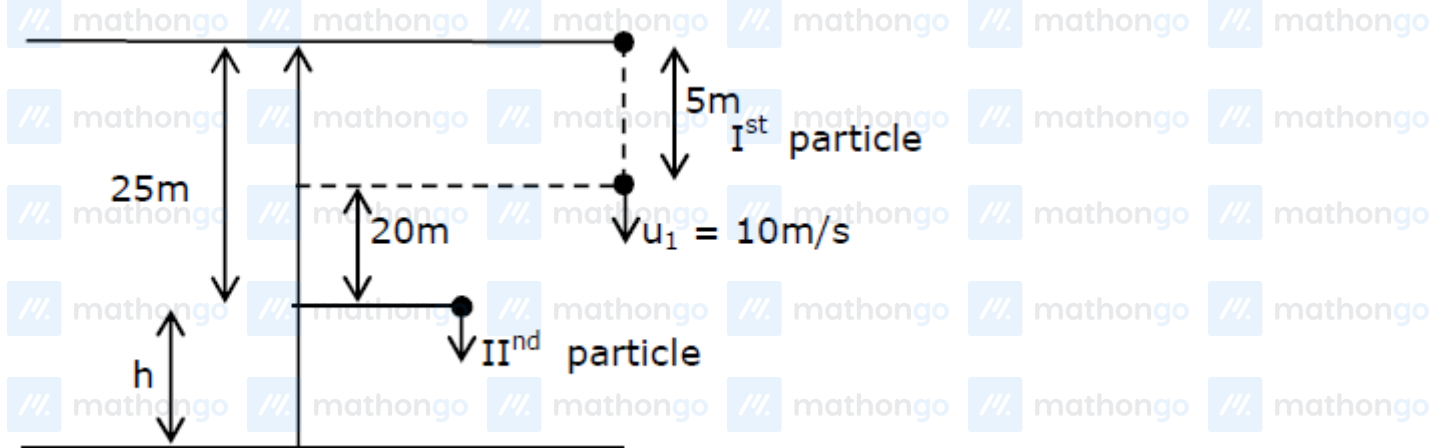
$$\therefore v_{\text{middle}}^2 = u^2 + al \quad \dots (2)$$

from equation (1) and (2) $v_{\text{middle}}^2 = u^2 + \left(\frac{v^2 - u^2}{2} \right)$

$$= \frac{v^2 + u^2}{2}$$

$$\therefore v_{\text{middle}} = \sqrt{\frac{v^2 + u^2}{2}}$$

Q3 (1)



For particle (1) $20 + h = 10t + \frac{1}{2}gt^2$ (i)

For particle (2) $h = \frac{1}{2}gt^2$ (ii)

put equation (ii) in equation (i)

$$20 + \frac{1}{2}gt^2 = 10t + \frac{1}{2}gt^2$$

Hints and Solutions

MathonGo

$$t = 2 \text{ sec}$$

Put in equation (ii)

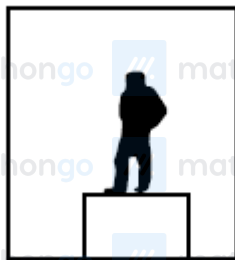
$$h = \frac{1}{2}gt^2$$

$$= \frac{1}{2} \times 10 \times 2^2$$

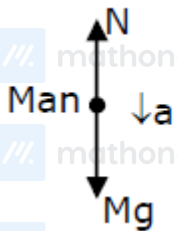
$$h = 20 \text{ m}$$

the height of the building = $25 + 20 = 45 \text{ m}$

Q4 (492)



$\downarrow 1.8 \text{ m/s}^2$



$$Mg - N = Ma$$

$$N = M(g - a)$$

$$N = 60(10 - 1.8)$$

$$N = 60 \times 8.2 = 492 \text{ N}$$

$$N = 492$$

Q5 (2)

From given information:

For 1st interval

$$a_1 = \frac{v_0}{t_1}$$

Hints and Solutions

MathonGo

$$v_0 = a_1 t_1 \quad \dots\dots (1)$$

For 2nd interval

$$a_2 = \frac{v_0}{t_2}$$

$$v_0 = a_2 t_2 \quad \dots\dots (2)$$

from (1)&(2)

$$a_1 t_1 = a_2 t_2$$

$$\frac{t_1}{t_2} = \frac{a_2}{a_1}$$

